Physics H Review Practice

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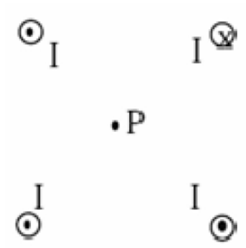
Final Range: Chapters 18-20

Note: (Review Guide/Review Practices/Practice Set) are UNOFFICIAL sources of reference and do NOT include or hint at questions in the actual exam!

Multiple Choice Practice

1. Four infinitely long wires are arranged as shown in the accompanying figure (end-on view). All four wires are perpendicular to the plane of the page and have the same magnitude of current *I*. Point P is a distance *a* from all four wires. What is the total magnetic field at point P?



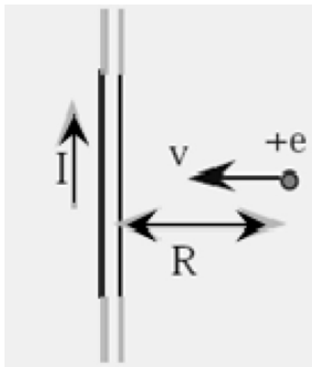
A)$\frac{\mu 0}{2\pi}\frac{I}{a}$ toward the upper left-hand corner

B)$\frac{\mu 0}{2\pi}\frac{I}{a}$ toward the lower left-hand corner

C)$2\frac{\mu 0}{2\pi}\frac{I}{a}$ toward the upper left-hand comer

D)0

2.The conventional current I in a long straight wire flows in the upward direction as shown in the figure (electron flow is downward). At the instant a proton of charge +e is a distance R from the wire and heading directly toward it, the force on the proton is:



A)$\frac{\mu 0}{2\pi}\frac{I^2R}{R}$ upward (in the same direction as I)

B)$\frac{\mu 0}{2\pi}\frac{I^2R}{R}$ downward (in the opposite direction as I)

C)$ev\frac{\mu 0}{2\pi}\frac{I^2R}{R}$ upward (in the same direction as I)

D)$ev\frac{\mu 0}{2\pi}\frac{I^2R}{R}$ downward (in the opposite direction as I)

3. A charged particle with constant speed enters a uniform magnetic field whose direction is perpendicular to the particle's velocity. The particle will:

A) Speed up

B) Experience no change in velocity

C) Follow a parabolic are

D) Follow a circular arc

4. A long straight wire conductor has constant and stable conventional current flowing towards the north direction. A compass is placed directly above the wire. When a large conventional current flows in the conductor as shown, the N pole of the compass:

A) has its polarity reversed

B) points to the south

C) points to the west

D)points to the east

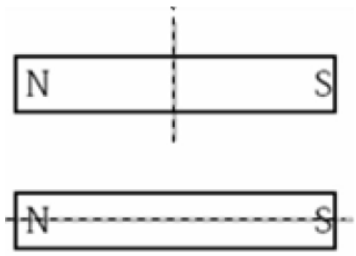
5. Two bar magnets are to be cut in half along the dotted lines shown. None of the pieces are rotated. After the cut:

A) The two halves of each magnet will attract each other

B) The two halves of each magnet will repel each other

C) The two halves of the top magnet will repel, the two halves of the bottom magnet will attract

D)The two halves of the top magnet will attract, the two halves of the bottom magnet will repel



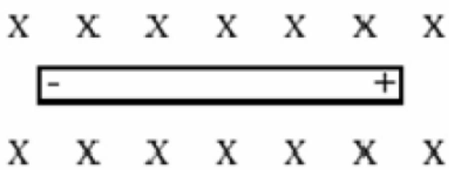
6. A wire moves through a magnetic field directed into the page. The wire experiences an induced charge separation as shown. Which way is the wire moving?

A) to the right

B) toward the top of the page

C) to the left

D) toward the bottom of the page



7. A charged particle with constant velocity enters a uniform magnetic field whose direction is parallel to the particle's velocity. The particle will

A) speed up

B) slow down

C) experience no change in velocity

D) follow a circular arc

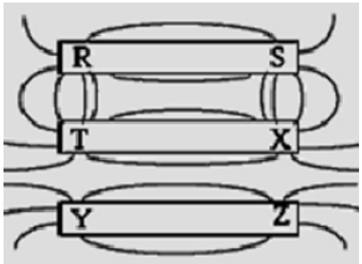
8. The diagram to the right depicts iron filings sprinkled around three permanent magnets. Pole R is the same pole as

A) T and Y

B) T and Z

C) X and Y

D) X and Z



9. Two light conductor wires are hung vertically. With electrical current in both wires directed upwards. Which of the following phenomenon occur?

A) the wires will experience a force of attraction

B) the wires will experience a force of repulsion

C) the force on the right-hand wire will cancel the force on the left-hand wire

D) both wires will experience a torque until they are at right angles to each other

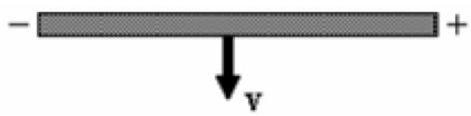
10. A wire moves with a velocity v through a magnetic field and experiences an induced charge separation as shown. What is the direction of the magnetic field?

A) into the page

B) towards the bottom of the page

C) out of the page

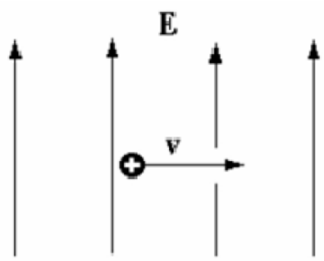
D) towards the top of the page



11. A positively charged particle moves to the right. It enters a region of space in which there is an electric field directed up the plane of the paper as shown. In which direction does the magnetic field have to point in this region so that the particle maintains a constant velocity?

A) into the plane of the page

B) out of the plane of the page



C)to the right

D) to the left

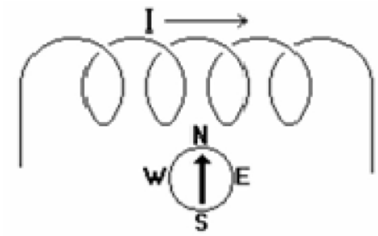
12. A compass is placed outside of a coil of wire (could be approximated as an unideal solenoid). A conventional electrical current is then run through the coil from left to right as shown. This will cause the North pole of the compass to:

A) point toward the left

B) point toward the right

C) point toward the bottom of the paper

D) not move since the magnetic field of the coil is into the paper



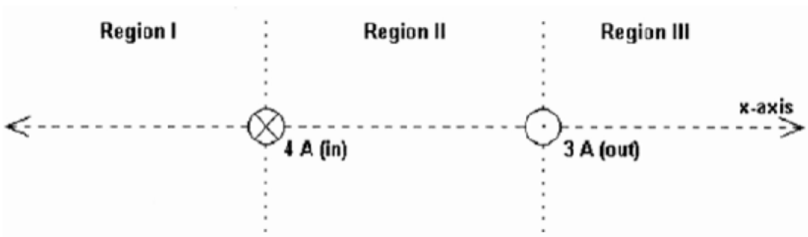
13. Two very long current–carrying wires are shown end on in the figure. The wire on the left has a 4A current going into the plane of the paper and the wire on the right has a 3A current coming out of the paper. Disregarding the case of x->∞, in which region(s) could the magnetic field from these two wires add to zero on the x–axis.

A) Region I only

B) Region II only

C) Region III only

D) Regions I and III only



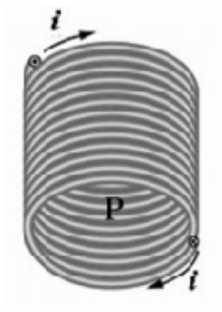
14.The magnetic field line passing through point P inside the solenoid is directed

A)to the right

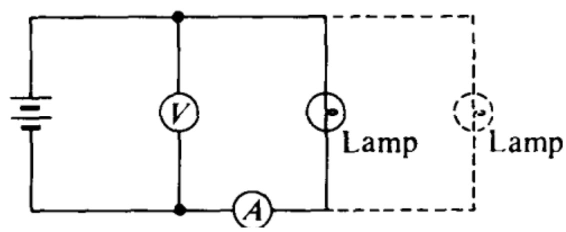
B)to the left

C)downward toward the bottom of the page

D)upward toward the top of the page



15.A lamp, a voltmeter V, an ammeter A, and a battery with zero internal resistance are connected as shown.



How would the ammeter's reading change when another lamp is connected in parallel with the first lamp as shown by the dashed lines?

(A)decreases, because the current through the ammeter splits to feed both branches

(B)remains the same, because the ammeter measures the current provided by the battery

(C)increases, because the net resistance of the circuit decreases

(D)remains the same, because energy is conserved in the circuit

16.How would the voltmeter's reading change when another lamp is connected in parallel with the first lamp as shown by the dashed lines?

(A)decreases, because the current is split between the two branches

(B)remains the same, because charge is conserved in the circuit

(C)increases, because the resistance of the circuit is increased

(D)remains the same, because the voltmeter is connected to the two ends of the battery

True/False Questions

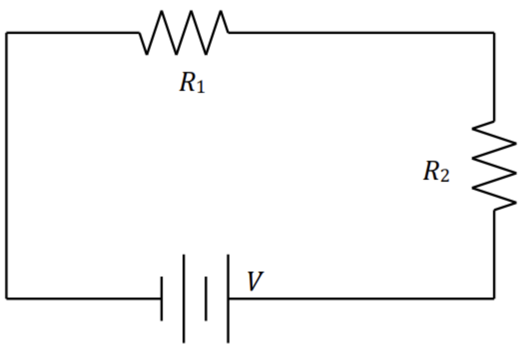
1. The right-hand rule can be used to determine the direction of the magnetic field inside a solenoid when the current direction is known. _________________

2. The magnetic force on a current-carrying wire is zero if the wire is parallel to the magnetic field. _________________

3. The magnetic field strength inside a solenoid increases if the number of turns per unit length decreases. _________________

4. A charged particle at rest in a uniform magnetic field experiences a magnetic force. _________________

Calculation



1.Two resistors are placed in series with a battery of potential V as shown above.

a.On the diagram, draw wires and a "**V**" to show where you would connect a voltmeter to measure the potential across R_1. If any existing wires need to be disconnected in the circuit, draw an X over those wires in the diagram.

b.On the diagram, draw wires and a "**A**" to show where you would connect an ammeter to measure the current traveling through R_2. If any existing wires need to be disconnected in the circuit, draw an X over those wires in the diagram.

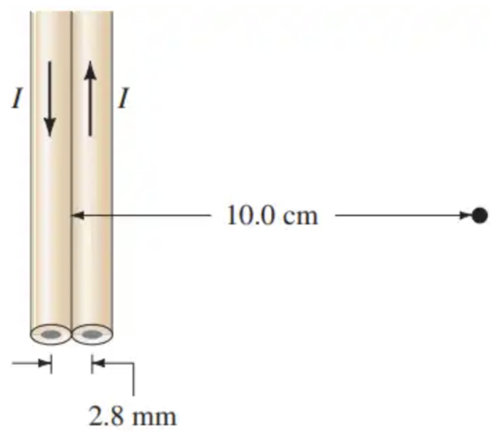
Resistance R_2 has a value of $1.00 \times 10^3\ \Omega$. The voltmeter shows the potential across R_1 to be 3.00V and the ammeter shows the current to be 5.00 mA through R_2.

c.Calculate the value of R_1.

d.Calculate the power dissipated by R_2.

e.Calculate the terminal voltage V of the battery.

f.The battery is disconnected from the circuit, and its potential measured to be 8.20 V. Determine the internal resistance of the battery when it was connected in the circuit.



2.A long pair of insulated wires serves to conduct 24.5 A of dc current to and from an instrument. If the wires are of negligible diameter but are 2.8 mm apart, what is the magnetic field 10.0 cm from their midpoint, in their plane? Compare to the magnetic field of the Earth.

3.A third wire is placed in the plane of the two wires shown in **(same figure)** parallel and just to the right. If it carries 25.0 A upward, what force per meter of length does it exert on each of the other two wires? Assume it is 2.8 mm from the nearest wire, center to center.